🏛 GLA UNIVERSITY, MATHURA — DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

HPC Masterclass | Day 5 In-Class Assessment: Generative AI, LLMs, Intel Gaudi & OpenVINO

Target Audience: 7th Semester B.Tech CSE (HPC Elective) | **Total Questions**: 25 | **Duration**: 30 Minutes

📋 Answer Key & Forensic Solution Guide (Instructor Copy)

Q#	Correct Answer	Cognitive Bloom Level	Core Concept Focus
01	B	`Understand`	In Large Language Model (LLM) inference on modern ...
02	B	`Knowledge`	What is the primary microarchitectural purpose of ...
03	B	`Analyze`	In the context of the Habana Gaudi 2 and Gaudi 3 d...
04	B	`Knowledge`	What two primary artifacts constitute the OpenVINO...
05	B	`Evaluate`	When applying Weight-Only INT4 Quantization using ...
06	B	`Apply`	In the mathematical formulation of Low-Rank Adapta...
07	C	`Apply`	Calculate the theoretical KV-cache memory required...
08	B	`Understand`	What is the primary role of the Habana SynapseAI G...
09	B	`Apply`	Which specific OpenVINO execution plugin should be...
10	B	`Analyze`	What is the key functional difference between Cont...
11	B	`Knowledge`	In the Slurm workload manager, what does the direc...
12	A	`Understand`	Which of the following quantization algorithms sup...
13	B	`Analyze`	In Habana Gaudi architecture, what is the role of ...
14	B	`Understand`	Why is PagedAttention (as implemented in modern LL...
15	B	`Knowledge`	What command-line tool in the Habana Gaudi Synapse...
16	B	`Apply`	When serving an OpenVINO LLM pipeline via a FastAP...
17	B	`Understand`	What does 'Group Size' (e.g., `group_size = 128`) ...
18	B	`Analyze`	In a distributed Gaudi cluster running DeepSpeed o...
19	B	`Knowledge`	What is the primary function of the OpenVINO Model...
20	B	`Apply`	If an LLM running on an Intel Xeon CPU achieves a ...
21	B	`Understand`	In QLoRA (Quantized Low-Rank Adaptation), what inn...

Q#	Correct Answer	Cognitive Bloom Level	Core Concept Focus
22	B	`Apply`	What is the primary purpose of the `compile_and_ru...
23	B	`Analyze`	How does Intel OpenVINO runtime achieve multi-sock...
24	B	`Knowledge`	When configuring a multi-node Gaudi training job o...
25	B	`Evaluate`	Why is INT4 weight-only quantization superior to I...

🔬 Complete Questions with Detailed Technical Rationale

Question 01

In Large Language Model (LLM) inference on modern server processors, why is the autoregressive token generation phase ('decode phase') classified as strictly memory-bandwidth bound rather than compute-bound?

- **(a)** Because the CPU halts its arithmetic logic units (ALUs) during token emission.
- **(b) [CORRECT]** Because generating each new token requires streaming billions of model weight parameters and past KV-cache states from DRAM to on-chip caches for just a single batch of matrix-vector multiplications, resulting in an operational arithmetic intensity far below the machine's ridge point.
- **(c)** Because matrix multiplications cannot be performed on floating-point numbers.
- **(d)** Because the operating system requires an Ethernet handshake before each emitted token.

> **Correct Answer**: **(B)**
> **Bloom's Cognitive Level**: `Understand`
> **Architectural Rationale**: During the autoregressive decode phase, prompt tokens have already been processed; each subsequent token is generated by loading the entire model weights (e.g., 14 GB for a 7B FP16 model) and KV-cache tensors just to compute one vector-matrix product per layer. With arithmetic intensity ~1-2 FLOPs/byte, DRAM bandwidth saturates long before compute engines reach peak utilization.

Question 02

What is the primary microarchitectural purpose of Intel Advanced Matrix Extensions (AMX) introduced in 4th/5th Gen Intel Xeon Scalable processors (Sapphire Rapids / Emerald Rapids)?

- **(a)** To replace DDR5 memory controllers with optical transceivers.
- **(b) [CORRECT]** To provide dedicated hardware 2D register tiles (TMM0-TMM7) and a systolic Tile Matrix Multiply Unit (TMUL) capable of performing 2,048 INT8 or 1,024 BF16 operations per cycle per core.
- **(c)** To convert Python bytecode into C++ at compile time.
- **(d)** To emulate ARM NEON instructions on x86 servers.

> **Correct Answer**: **(B)**
> **Bloom's Cognitive Level**: `Knowledge`
> **Architectural Rationale**: Intel AMX consists of two main architectural components: eight 1KB 2D tile registers (TILECFG / TMM0-TMM7) and a systolic matrix execution unit called TMUL. TMUL performs bfloat16 (TDPBF16PS) and INT8 (TDPBUSD) dot-product matrix multiplies, delivering up to an 8x throughput leap over AVX-512 VNNI.

Question 03

In the context of the Habana Gaudi 2 and Gaudi 3 deep learning accelerators, how is scale-out inter-node communication uniquely implemented compared to traditional GPU architectures?

- **(a)** Through standard USB-C external dongles connected to each server chassis.
- **(b) [CORRECT]** By integrating 24 standard 100GbE (Gaudi 2) or 200GbE (Gaudi 3) RDMA over Converged Ethernet (RoCE v2) network ports directly onto each accelerator die, eliminating external PCIe NIC bottlenecks and vendor-proprietary switch lock-in.
- **(c)** Using optical fiber cables plugged directly into the CPU L3 cache controller.
- **(d)** By writing all intermediate tensors to local SATA hard drives for cluster exchange.

> **Correct Answer**: **(B)**
> **Bloom's Cognitive Level**: `Analyze`
> **Architectural Rationale**: Habana Gaudi eliminates external PCIe Host Channel Adapters (HCAs) by integrating 24 RoCE v2 engines natively on-chip. In an 8-card subsystem, 21 ports are dedicated to an all-to-all internal mesh, while 3 ports per card (24x 100GbE / 200GbE ports per server) directly drive standard spine-leaf Ethernet fabrics for linear multi-node scaling.

Question 04

What two primary artifacts constitute the OpenVINO Intermediate Representation (IR) format produced when compiling a deep learning model?

- **(a)** A `.zip` archive containing Python source code and a Dockerfile.
- **(b) [CORRECT]** An XML file (`.xml`) describing the network topology and computational graph, accompanied by a binary file (`.bin`) storing the optimized tensor weights and biases.
- **(c)** A `.json` configuration file and a `.tar.gz` Linux kernel image.
- **(d)** A C++ header file (`.h`) and an executable binary (`.elf`).

> **Correct Answer**: **(B)**
> **Bloom's Cognitive Level**: `Knowledge`
> **Architectural Rationale**: The OpenVINO IR representation bifurcates the model into two complementary files: the `.xml` file (human-readable XML specifying layer connections, shapes, precisions, and operator types) and the `.bin` file (raw contiguous binary memory storing trained weights, scale factors, and biases).

Question 05

When applying Weight-Only INT4 Quantization using Intel NNCF (Neural Network Compression Framework) on an LLM, why are the attention projections and MLP weights quantized to INT4 while activations remain in FP16 / BF16?

- **(a)** Because INT4 activations are illegal under IEEE floating-point standards.
- **(b) [CORRECT]** Because quantizing activations dynamically requires complex runtime calibration and can severely degrade model perplexity, whereas quantizing static weights alone cuts DRAM bandwidth consumption by ~4x with negligible loss in accuracy during memory-bound decoding.
- **(c)** Because OpenVINO cannot perform integer multiplication.
- **(d)** Because NNCF only supports convolution layers, not dense layers.

> **Correct Answer**: **(B)**
> **Bloom's Cognitive Level**: `Evaluate`
> **Architectural Rationale**: In LLM autoregressive decoding, memory bandwidth is the primary bottleneck. Weight-Only INT4 quantization shrinks weight footprints from 16 bits to 4 bits per parameter, slashing memory traffic by ~75%. Activations (token representations) remain in FP16/BF16, avoiding dynamic outlier clipping and preserving generational quality.

Question 06

In the mathematical formulation of Low-Rank Adaptation (LoRA), given a frozen weight matrix $W_0 \in \mathbb{R}^{d \times k}$, how is the updated weight matrix W parameterized during fine-tuning?

- **(a)** $W = W_0 \times (B + A)$
- **(b) [CORRECT]** $W = W_0 + \Delta W = W_0 + \frac{\alpha}{r}(B \cdot A)$, where $B \in \mathbb{R}^{d \times r}$, $A \in \mathbb{R}^{r \times k}$, and rank $r \ll \min(d, k)$.
- **(c)** $W = \frac{W_0}{\alpha \cdot r} + \log(B \cdot A)$
- **(d)** $W = W_0 - (A^T \cdot B^T)$

> **Correct Answer**: **(B)**
> **Bloom's Cognitive Level**: `Apply`
> **Architectural Rationale**: LoRA freezes the pre-trained weight matrix W_0 and decomposes the task-specific update ΔW into the product of two low-rank matrices: $B \in \mathbb{R}^{d \times r}$ (initialized to zeros) and $A \in \mathbb{R}^{r \times k}$ (initialized with Gaussian noise). The hyperparameter α scales the update, drastically reducing trainable parameters by up to 99.9%.

Question 07

Calculate the theoretical KV-cache memory required for a 7-billion parameter transformer model with 32 layers, 32 attention heads, hidden dimension 4,096 ($d_{head} = 128$), running at FP16 (2 bytes per element), with a batch size of 8 and a context sequence length of 2,048 tokens.

- **(a)** 128 MB
- **(b)** 1.0 GB
- **(c) [CORRECT]** 2.15 GB (approximately 2.0×10^9 bytes)
- **(d)** 64.0 GB

> **Correct Answer**: **(C)**
> **Bloom's Cognitive Level**: `Apply`
> **Architectural Rationale**: KV-cache formula: $2 \times \text{Layers} \times \text{Heads} \times d_{head} \times \text{SeqLen} \times \text{Batch} \times \text{BytesPerElem}$. Here: $2 \times 32 \times 32 \times 128 \times 2048 \times 8 \times 2 = 2 \times 32 \times 4096 \times 2048 \times 8 \times 2 = 8,589,934,592 \text{ bytes}/4 = 2,147,483,648 \text{ bytes} = 2.0 \text{ GiB} \approx 2.15 \text{ GB}$.

Question 08

What is the primary role of the Habana SynapseAI Graph Compiler when executing PyTorch code on Gaudi HPUs via Lazy Mode (`PT_HPU_LAZY_MODE=1`)?

- **(a)** It translates Python syntax directly into HTML web documentation.

- **(b) [CORRECT]** It traces PyTorch operations across multiple execution steps, constructs an optimized computational Directed Acyclic Graph (DAG), fuses compatible tensor kernels, schedules DMA transfers, and dispatches compiled recipes to the HPU engines with minimal host CPU intervention.
- **(c)** It forces PyTorch to execute in eager mode with line-by-line CPU synchronization.
- **(d)** It downsamples image datasets to thumbnail resolution before training.

> **Correct Answer**: **(B)**
> **Bloom's Cognitive Level**: `Understand`
> **Architectural Rationale**: SynapseAI Lazy Mode intercepts PyTorch operations into an intermediate computational graph rather than executing them eagerly. The Graph Compiler then performs global memory allocation, operator fusion (e.g., fusing MatMul with bias addition and activation), and generates an optimized execution recipe for the Gaudi MME and TPC engines.

Question 09

Which specific OpenVINO execution plugin should be targeted to leverage Intel AMX acceleration on 4th or 5th Gen Intel Xeon Scalable processors?

- **(a)** `core.compile_model(model, 'GPU')`
- **(b) [CORRECT]** `core.compile_model(model, 'CPU')` with execution precision set to `ov::element::bf16` or `ov::element::i8`.
- **(c)** `core.compile_model(model, 'NPU_V1')`
- **(d)** `core.compile_model(model, 'FPGA_BITSTREAM')`

> **Correct Answer**: **(B)**
> **Bloom's Cognitive Level**: `Apply`
> **Architectural Rationale**: The OpenVINO `CPU` plugin dynamically detects on-chip host capabilities via CPUID. When running on Sapphire Rapids or Emerald Rapids with BF16 or INT8 precision enabled, the CPU plugin automatically targets the AMX TMUL hardware tiles for matrix multiplications.

Question 10

What is the key functional difference between Continuous Batching (also known as Iteration-Level Batching) and traditional Static Batching in LLM serving engines?

- **(a)** Continuous batching requires all requests in a batch to have identical prompt lengths and generation lengths.
- **(b) [CORRECT]** Continuous batching dynamically inserts newly arriving requests into the active iteration loop and evicts finished requests immediately upon emitting an EOS token, eliminating padding overhead and GPU idle time.
- **(c)** Continuous batching runs only on single-core microcontrollers.
- **(d)** Static batching generates tokens 10x faster than continuous batching.

> **Correct Answer**: **(B)**
> **Bloom's Cognitive Level**: `Analyze`
> **Architectural Rationale**: In static batching, all requests in a batch are padded to the maximum sequence length and the entire batch must wait until the slowest request finishes before new work can be scheduled. Continuous batching operates at the token-iteration level, retiring completed sequences and admitting new requests dynamically at every single decode step.

Question 11

In the Slurm workload manager, what does the directive `#SBATCH --gres=habana:8` specify when submitting a distributed training job?

- **(a)** It requests 8 gigabytes of RAM from the cluster master node.
- **(b) [CORRECT]** It requests 8 generic accelerator resources (Habana Gaudi HPUs) allocated exclusively to the job on each assigned compute node.
- **(c)** It forces Slurm to rerun the job 8 times upon failure.
- **(d)** It sets the network timeout threshold to 8 minutes.

> **Correct Answer**: **(B)**
> **Bloom's Cognitive Level**: `Knowledge`
> **Architectural Rationale**: Slurm uses the Generic Resource (`--gres`) flag to manage heterogeneous accelerators. `--gres=habana:8` tells Slurm to schedule a node containing 8 Habana Gaudi HPU accelerator devices and assign them exclusively to the submitting job's cgroup.

Question 12

Which of the following quantization algorithms supported by Intel NNCF selectively identifies salient weight channels that correspond to large activation outliers and preserves their precision to maintain generative accuracy in INT4?

- **(a) [CORRECT]** AWQ (Activation-aware Weight Quantization)
- **(b)** Random Uniform Dithering
- **(c)** Binary Thresholding
- **(d)** Zero-Order Linear Extrapolation

> **Correct Answer**: **(A)**
> **Bloom's Cognitive Level**: `Understand`
> **Architectural Rationale**: AWQ recognizes that not all weights are equally important: protecting the top 1% of salient weight channels (those multiplied by high-magnitude activation features) from aggressive quantization error prevents perplexity degradation while quantizing the remaining 99% of weights to INT4.

Question 13

In Habana Gaudi architecture, what is the role of the Matrix Multiplication Engine (MME) versus the Tensor Processor Core (TPC)?

- **(a)** MME handles disk I/O, while TPC renders 3D graphics.
- **(b) [CORRECT]** MME is a dedicated fixed-function systolic engine optimized for dense matrix multiplications and convolutions, while TPC is a fully programmable VLIW SIMD processor handling activations, normalizations, and custom non-linear tensor ops.
- **(c)** MME handles network packets; TPC manages CPU cooling fans.
- **(d)** TPC performs matrix multiplications, while MME compiles Python scripts.

> **Correct Answer**: **(B)**
> **Bloom's Cognitive Level**: `Analyze`
> **Architectural Rationale**: Gaudi separates compute workloads between two specialized silicon blocks: the MME performs massive 2D matrix multiplications (GEMMs) with maximum energy efficiency, while TPCs (programmable via LLVM-based C/C++ kernels) execute element-wise operations, Softmax, LayerNorm, and custom activations.

Question 14

Why is PagedAttention (as implemented in modern LLM serving engines) considered an architectural breakthrough for KV-cache memory management?

- **(a)** It eliminates the need for any KV-cache in transformer models.
- **(b) [CORRECT]** It borrows the classic OS virtual memory paging concept to store KV-cache tensors in non-contiguous physical memory blocks, completely eliminating internal and external memory fragmentation and enabling near-zero memory waste (~4%).
- **(c)** It writes all attention matrices directly to NVMe SSDs.
- **(d)** It doubles the parameter count of the underlying neural network.

> **Correct Answer**: **(B)**
> **Bloom's Cognitive Level**: `Understand`
> **Architectural Rationale**: Traditional KV-cache allocation pre-allocates contiguous memory blocks for the maximum possible sequence length (e.g., 4K tokens), leading to 60-80% memory waste due to internal and external fragmentation. PagedAttention divides the KV-cache into discrete physical pages (e.g., 16 tokens/page) linked via a page table, allowing dynamic allocation.

Question 15

What command-line tool in the Habana Gaudi SynapseAI software stack provides real-time monitoring of HPU temperature, memory consumption, engine utilization, and power draw?

- **(a)** `nvidia-smi`
- **(b) [CORRECT]** `hl-smi`
- **(c)** `rocm-smi`
- **(d)** `top-hpc`

> **Correct Answer**: **(B)**
> **Bloom's Cognitive Level**: `Knowledge`
> **Architectural Rationale**: The Habana Labs System Management Interface (`hl-smi`) is the official utility for querying and monitoring Habana Gaudi HPU accelerators, displaying PCIe/HBM status, board temperatures, power draw, and per-process HPU memory allocations.

Question 16

When serving an OpenVINO LLM pipeline via a FastAPI microservice, what is the key architectural advantage of returning an asynchronous `StreamingResponse` using an `AsyncIterator` over returning a standard JSON response?

- **(a)** It encrypts the payload using AES-256 before transmission.

- **(b) [CORRECT]** It yields generated tokens over an HTTP/1.1 chunked transfer stream immediately as they are decoded, drastically reducing Time-to-First-Token (TTFT) perceived latency from seconds to milliseconds for end users.
- **(c)** It forces the client browser to refresh the webpage automatically.
- **(d)** It converts text tokens into WAV audio files.

> **Correct Answer**: **(B)**
> **Bloom's Cognitive Level**: `Apply`
> **Architectural Rationale**: A standard JSON response requires the server to wait until the entire sequence (e.g. 500 tokens) finishes generating before sending the HTTP payload. An asynchronous `StreamingResponse` streams each token over Server-Sent Events (SSE) or chunked transfer encoding as soon as it is emitted by the OpenVINO tokenizer, yielding sub-100ms perceived latency.

Question 17

What does 'Group Size' (e.g., `group_size = 128`) mean when configuring INT4 weight quantization in NNCF or GPTQ?

- **(a)** The number of students working together on a single HPC node.
- **(b) [CORRECT]** The number of consecutive weight elements along a channel that share a single floating-point scale factor and zero-point offset.
- **(c)** The number of CPU sockets installed on the server motherboard.
- **(d)** The total count of attention layers in the transformer model.

> **Correct Answer**: **(B)**
> **Bloom's Cognitive Level**: `Understand`
> **Architectural Rationale**: Per-tensor quantization uses one scale factor for the entire tensor, while per-channel quantization uses one scale per row/column. Group quantization breaks channels into smaller blocks (typically 64 or 128 elements), assigning independent scale and zero-point parameters to each group to better capture local dynamic ranges.

Question 18

In a distributed Gaudi cluster running DeepSpeed or PyTorch FSDP (Fully Sharded Data Parallel), what does 'ZeRO Stage 3' shard across all cluster nodes?

- **(a)** Only the Python script source code files.
- **(b) [CORRECT]** Optimizer states, gradients, and model parameters (weights), so that each accelerator only stores $1/N$ of the total model state during forward and backward passes.
- **(c)** The Linux operating system kernel image.
- **(d)** Only the validation dataset images.

> **Correct Answer**: **(B)**
> **Bloom's Cognitive Level**: `Analyze`
> **Architectural Rationale**: ZeRO (Zero Redundancy Optimizer) Stage 1 shards optimizer states (4x memory reduction); Stage 2 shards optimizer states and gradients (8x memory reduction); Stage 3 shards optimizer states, gradients, AND model parameters, reducing memory footprint linearly with cluster size ($1/N$).

Question 19

What is the primary function of the OpenVINO Model Optimizer / `optimum-cli export openvino` tool when ingesting a Hugging Face PyTorch model?

- **(a)** It deletes all layers containing linear transformations to save disk space.
- **(b) [CORRECT]** It traces the PyTorch model graph, performs hardware-agnostic optimizations (such as dead-code elimination, constant folding, and operator fusion), and serializes the network into the OpenVINO IR format (`.xml` and `.bin`).
- **(c)** It automatically generates a PowerPoint presentation explaining the model.
- **(d)** It uploads the model to a public FTP server.

> **Correct Answer**: **(B)**
> **Bloom's Cognitive Level**: `Knowledge`
> **Architectural Rationale**: `optimum-cli export openvino` or OpenVINO's Model Converter ingests PyTorch/ONNX models, identifies sub-graphs suitable for fusion (e.g. GELU, RMSNorm), resolves dynamic dimensions, and exports the optimized static/dynamic computational graph to IR.

Question 20

If an LLM running on an Intel Xeon CPU achieves a Time-to-First-Token (TTFT) of 150 ms and an Inter-Token Latency (ITL) of 25 ms, what is its generation throughput in Tokens Per Second (TPS) during the decode phase?

- **(a)** 4.0 TPS
- **(b) [CORRECT]** 40.0 TPS
- **(c)** 6.67 TPS
- **(d)** 150.0 TPS

> **Correct Answer**: **(B)**
> **Bloom's Cognitive Level**: `Apply`
> **Architectural Rationale**: Generation throughput during the decode phase is inversely proportional to the Inter-Token Latency: $\text{TPS} = \frac{1}{\text{ITL (seconds)}} = \frac{1}{0.025\text{ s}} = 40.0\ \text{tokens/second}$.

Question 21

In QLoRA (Quantized Low-Rank Adaptation), what innovative 4-bit data type is introduced to quantize the frozen base model weights without sacrificing zero-mean normal distribution fidelity?

- **(a)** IEEE 754 Single Precision
- **(b) [CORRECT]** NormalFloat 4 (NF4)
- **(c)** Standard BCD (Binary Coded Decimal)
- **(d)** Signed 4-bit Twos Complement Integer

> **Correct Answer**: **(B)**
> **Bloom's Cognitive Level**: `Understand`
> **Architectural Rationale**: NF4 (NormalFloat 4) is an information-theoretically optimal quantile quantization data type designed for normally distributed neural network weights, ensuring that each of the 16 quantization bins has an equal number of expected weight parameters.

Question 22

What is the primary purpose of the `compile_and_run_all.sh` test harness script in an HPC lab environment?

- **(a)** To wipe the cluster file system and restore factory settings.
- **(b) [CORRECT]** To systematically validate environment dependencies, compile native/Python code, execute benchmarking workloads, and output rigorous PASS/FAIL assertions with execution timestamps.
- **(c)** To mine cryptocurrencies during idle server hours.
- **(d)** To shut down the server power supply safely.

> **Correct Answer**: **(B)**
> **Bloom's Cognitive Level**: `Apply`
> **Architectural Rationale**: An automated test harness systematically walks through the lab's artifacts, verifying runtime prerequisites, checking script syntax, executing integration pipelines, and producing reproducible logs for student grading and infrastructure verification.

Question 23

How does Intel OpenVINO runtime achieve multi-socket CPU scaling on dual-socket 4th Gen Intel Xeon servers?

- **(a)** By turning off NUMA support in the server BIOS.
- **(b) [CORRECT]** By partitioning threads into NUMA-aware execution streams (`ov::num_streams`), binding inference worker streams to localized CPU cores and memory nodes to avoid costly cross-socket UPI interconnect traffic.
- **(c)** By running one copy of Windows and one copy of Linux concurrently.
- **(d)** By duplicating all model weights across CPU L1 caches.

> **Correct Answer**: **(B)**
> **Bloom's Cognitive Level**: `Analyze`
> **Architectural Rationale**: On NUMA systems, cross-socket memory access over UPI incurs a ~2-3x latency penalty. OpenVINO's CPU Streams architecture (`ov::hint::num_streams`) pins independent worker threads to specific NUMA nodes and memory pools, maximizing aggregate parallel throughput.

Question 24

When configuring a multi-node Gaudi training job on Slurm, which environment variable ensures that PyTorch communications use Habana Collective Communications Library (HCCL) over RoCE interfaces?

- **(a)** `NCCL_DEBUG=INFO`
- **(b) [CORRECT]** `HCCL_OVER_OFI=1` and `PT_HPU_ENABLE_LAZY_COLLECTIVES=true`
- **(c)** `CUDA_VISIBLE_DEVICES=none`
- **(d)** `MPI_BYPASS_ALL=1`

> **Correct Answer**: **(B)**
> **Bloom's Cognitive Level**: `Knowledge`
> **Architectural Rationale**: Habana uses HCCL (Habana Collective Communications Library) as its native collective engine. Setting `HCCL_OVER_OFI=1` directs HCCL to utilize Open Fabrics Interfaces (OFI) to drive the on-die RoCE ports for distributed all-reduce operations across cluster nodes.

Question 25

Why is INT4 weight-only quantization superior to INT8 weight-only quantization for LLM serving on memory-bandwidth-limited edge and server CPUs?

- **(a)** Because INT4 requires zero CPU power consumption.
- **(b) [CORRECT]** Because INT4 cuts the weight tensor size in half compared to INT8 (and 4x compared to FP16), halving the bytes that must be read across the DDR5 memory bus per emitted token and translating directly to a near-2x boost in decode throughput.
- **(c)** Because INT4 models cannot produce hallucinations.
- **(d)** Because INT4 completely disables the softmax operator.

> **Correct Answer**: **(B)**
> **Bloom's Cognitive Level**: `Evaluate`
> **Architectural Rationale**: Since autoregressive decode is strictly memory-bandwidth bound, the generation speed is directly proportional to memory throughput divided by model size in bytes ($Tokens/sec \propto Bandwidth/ModelBytes$). Halving the model bytes from INT8 (1 byte/param) to INT4 (0.5 bytes/param) doubles the theoretical upper bound of decode speed.